

Suhwan Kim

professor

基本信息

姓名: Suhwan Kim

所属院校: 首尔国立大学

地区: South Korea

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

低功耗电路与能量管理

模拟与数模混合电路

数字集成电路设计

存储器设计与架构

混合信号集成技术

超大规模集成电路设计

模拟与射频IC设计

微电子器件与制造

射频集成电路

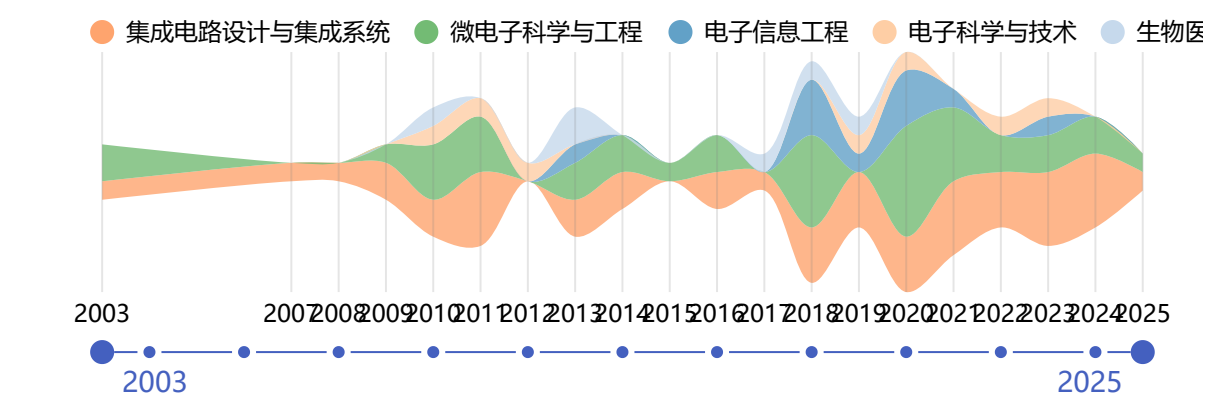
个人简介

Suhwan Kim is a Professor in the Department of Electrical and Computer Engineering at Seoul National University. His research focuses on low-power VLSI design, adiabatic circuits, and power gating structures. He has authored numerous publications in these areas and collaborates with various institutions and industry partners, including Kwangwoon University, ETRI, SK hynix, and Samsung Electronics. His work contributes significantly to advancements in digital systems and low-power circuit technology.

工作经历

Seoul National University - Professor

近年发表的论文



A 13.2-kSPS Data Rate, 5.1-G Ω Input Impedance Read-Out IC for DC Measurements
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS
Park, Junyoung; Kim, Wooyoung; Han, Donghoon; Lee, Hyunjoong; Kim, Suhwan; Rhee, Jooyeol
被引用次数: 0 2025

A 5.5-276 W Integrated Photovoltaic Energy Harvesting System With 99.95% MPPT Efficiency and 98.74% Power Conversion Efficiency
IEEE TRANSACTIONS ON POWER ELECTRONICS
Yang, Jinho; Lee, Hyunjoong; Cho, Jun Soo; Lim, Kyoungwon; Kim, Hwi-Cheol; Kim, Suhwan; Rhee, Jooyeol
被引用次数: 0 2025

A 0.77-pJ/bit 40-Gb/s/pin Single-Ended Hybrid DAC-Based Transmitter for Memory Interfaces
2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024
Seo, Sanghyuk; Jeong, Yong-un; Yun, Jeakwang; Kim, Jaewook; Kim, Suhwan
被引用次数: 0 2024

An N/PBTI-Isolated BTI Monitor With a Configurable Switching Network and Calibration for Process Variation in Memory Periphery
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS
Jeong, Shin-Hyun; Jeong, Yong-Un; Kim, Suhwan
被引用次数: 0 2024

A 10-bit 13.3 μ W single-slope analog-to-digital converter with auto-zero power-down technique
ELECTRONICS LETTERS
Lee, Youngwoo; Kim, Suhwan; Jun, Jaehoon
被引用次数: 0 2024

A Single-Ended Impedance-Matched Transmitter With Single Ring-Oscillator-Based Time-Domain ZQ Calibration for Memory Interfaces
IEEE JOURNAL OF SOLID-STATE CIRCUITS
Yun, Jaekwang; Lee, Sangyoon; Kim, Jaewook; Chae, Joo-Hyung; Kim, Suhwan; Jeong, Yong-Un

被引用次数: 0

2024

A 50-1600 MHz Wide-Range Digital Duty-Cycle Corrector With Counter-Based Half-Cycle Delay Line

IEEE ACCESS

Kim, Jaewook; Yun, Jaekwang; Chae, Joo-Hyung; Kim, Suhwan

被引用次数: 2

2023

A 105 dB Effective Dynamic Range Self- Capacitance Proximity Sensing System With Consecutive Double-Sided Conversion Technique

IEEE SENSORS JOURNAL

Kim, Minsung; Han, Sanghwa; Kim, Suhwan; Rhee, Jooyeol

被引用次数: 0

2023

Single-Ended Receiver-Side Crosstalk Cancellation With Independent Gain and Timing Control for Minimum Residual FEXT

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jeong, Yong-Un; Choi, Sungphil; Kim, Suhwan; Chae, Joo-Hyung

被引用次数: 4

2023

A 0.85-pJ/b 16-Gb/s/Pin Single-Ended Transmitter With Integrated Voltage Modulation for Low-Power Memory Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Yong-Un; Chae, Joo-Hyung; Kim, Suhwan

被引用次数: 2

2023

A 17.6-Bit 800-SPS Energy-Efficient Read-Out IC With Input Impedance Boosting

IEEE SENSORS JOURNAL

Park, Junyoung; Kim, Wooyoung; Kim, Daesik; Lee, Dongkeon; Hong, Yongtaek; Kim, Suhwan; Rhee, Jooyeol

被引用次数: 3

2023

A 19-bit Range and 4.5-ps Resolution Fully-Synthesizable Time-to-Digital Converter with Quad-Edge Offset Cancellation

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Cheong, Heon Hwa; Kim, Suhwan

被引用次数: 0

2022

A Low-Power DRAM Transmitter With Phase and Current-Mode Amplitude Equalization to Improve Impedance Matching

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Sangyoon; Jeong, Yong-Un; Yun, Jaekwang; Chae, Joo-Hyung; Kim, Suhwan

被引用次数: 2

2022

A 10 Gb/s/pin Single-Ended Transmitter With Reflection-Aided Duobinary Modulation for Dual-Rank Mobile Memory Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jeong, Yong-Un; Choi, Sungphil; Chae, Joo-Hyung; Yun, Jaekwang; Jeong, Shin-Hyun; Kim, Suhwan
被引用次数: 4 2022

A Low-Power Digital Capacitive MEMS Microphone Based on a Triple-Sampling Delta-Sigma ADC With Embedded Gain

IEEE ACCESS

Lee, Byunggyu; Yang, Jinho; Cho, Jun Soo; Kim, Suhwan

被引用次数: 5 2022

A 24-Gb/s/pin Single-Ended PAM-4 Receiver With 1-Tap Decision Feedback Equalizer Using Inverter-Based Summer for Memory Interfaces

IEEE ACCESS

Park, Hyunkyu; Jeong, Yong-Un; Kim, Suhwan

被引用次数: 2 2022

A 0.64-pJ/Bit 28-Gb/s/Pin High-Linearity Single-Ended PAM-4 Transmitter With an Impedance-Matched Driver and Three-Point ZQ Calibration for Memory Interface

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Yong-Un; Park, Hyunkyu; Hyun, Changho; Chae, Joo-Hyung; Jeong, Shin-Hyun; Kim, Suhwan

被引用次数: 26 2021

A $\pm 0.15\%$ RH Inaccuracy Humidity Sensing System With ± 0.44 °C (3σ) Inaccuracy On-Chip Temperature Sensor

IEEE SENSORS JOURNAL

Jun, Jaehoon; Shin, Sangmin; Kim, Minsung; Ahn, Yongjoon; Kim, Suhwan

被引用次数: 5 2021

Under-Display Optical Proximity Sensor With Silicon Nanowires

IEEE SENSORS JOURNAL

Rhee, Cyuyeol; Kim, Hyunjong; Yang, Han; Kim, Suhwan; Lee, Hyunjoong

被引用次数: 3 2021

A Controller PHY for Managed DRAM Solution With Damping-Resistor-Aided Pulse-Based Feed-Forward Equalizer

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ko, Hyeongjun; Kim, Mino; Park, Hyunkyu; Lee, Sangyoon; Kim, Jaewook; Kim, Suhwan; Chae, Joo-Hyung

被引用次数: 5 2021

A Differentiating Receiver With a Transition-Detecting DFE for Dual-Rank Mobile Memory Interface

IEEE ACCESS

Choi, Sungphil; Jeong, Yong-Un; Chae, Joo-Hyung; Jeong, Shin-Hyun; Kim, Suhwan

被引用次数: 2 2021

A Fast-Locking All-Digital PLL With Triple-Stage Phase-Shifting

IEEE ACCESS

Cheong, Heon Hwa; Kim, Suhwan

被引用次数: 4

2021

An 18-Gb/s/pin Single-Ended PAM-4 Transmitter for Memory Interfaces with Adaptive Impedance Matching and Output Level Compensation

ELECTRONICS

Hyun, Changho; Jeong, Yong-Un; Kim, Suhwan; Chae, Joo-Hyung

被引用次数: 2

2021

A 0.3 lx-1.4 Mlx Monolithic Silicon Nanowire Light-to-Digital Converter With Temperature-Independent Offset Cancellation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Rhee, Cyuyeol; Park, Junyoung; Kim, Suhwan

被引用次数: 8

2020

A Wide Dynamic Range Multi-Mode Band-Pass Continuous-Time Delta-Sigma Modulator Employing Single-Opamp Resonator With Positive Resistor-Feedback

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Susie; Rhee, Cyuyeol; Kim, Suhwan

被引用次数: 6

2020

A 10.4-Gb/s 1-Tap Decision Feedback Equalizer With Different Pull-Up and Pull-Down Tap Weights for Asymmetric Memory Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chae, Joo-Hyung; Kim, Minchang; Choi, Sungphil; Kim, Suhwan

被引用次数: 13

2020

A 9Gb/s Wide Output Range Transmitter With 2D Binary-Segmented Driver and Dual-Loop Calibration for Intra-Panel Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jeong, Yong-Un; Park, Jihwan; Kim, Mino; Chae, Joo-Hyung; Yun, Jaekwang; Lee, Hyunjoong; Kim, Suhwan

被引用次数: 3

2020

A Digital Capacitive MEMS Microphone for Speech Recognition With Fast Wake-Up Feature Using a Sound Activity Detector

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yang, Youngtae; Lee, Byunggyu; Cho, Jun Soo; Kim, Suhwan; Lee, Hyunjoong

被引用次数: 13

2020

A 0.45 pJ/b, 6.4 Gb/s Forwarded-Clock Receiver With DLL-Based Self-Tracking Loop for Unmatched Memory Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Shin, Soyeong; Ko, Han-Gon; Kye, Chan-Ho; Lee, Sang-Yoon; Yun, Jaekwang; Lee, Doobock; Jung, Hae-Kang; Kim, Suhwan; Jeong, Deog-Kyoon

被引用次数: 1

2020

A 0.83-pJ/Bit 6.4-Gb/s HBM Base Die Receiver Using a 45° Strobe Phase for Energy-Efficient Skew Compensation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Sangyoon; Ko, Han-Gon; Chae, Joo-Hyung; Shin, Soyeong; Yun, Jaekwang; Jeong, Deog-Kyoon; Kim, Suhwan

被引用次数: 2

2020

Level-shifting Low-pass Filter Using DC-decoupled Capacitance Multiplication for Integrated Input Interface of Automotive MCUs

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Hyunjong; Park, Yujin; Kim, Suhwan; Lee, Hyunjoong

被引用次数: 0

2020

Data-Dependent Selection of Amplitude and Phase Equalization in a Quarter-Rate Transmitter for Memory Interfaces

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chae, Joo-Hyung; Jeong, Yong-Un; Kim, Suhwan

被引用次数: 14

2020

A Quadrature Clock Corrector for DRAM Interfaces, With a Duty-Cycle and Quadrature Phase Detector Based on a Relaxation Oscillator

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chae, Joo-Hyung; Ko, Hyeongjun; Park, Jihwan; Kim, Suhwan

被引用次数: 21

2019

A 22-bit Read-Out IC With 7-ppm INL and Sub-100-μHz 1/f Corner for DC Measurement Systems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jun, Jaehoon; Park, Soungchul; Kang, Junho; Kim, Suhwan

被引用次数: 14

2019

An 80.2 dB DR 23.25 mW/Channel 8-Channel Ultrasound Receiver With a Beamforming Embedded SAR ADC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Taehoon; Shin, Sangmin; Kim, Suhwan

被引用次数: 18

2019

A 12.8-Gb/s Quarter-Rate Transmitter Using a 4:1 Overlapped Multiplexing Driver Combined With an Adaptive Clock Phase Aligner

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chae, Joo-Hyung; Ko, Hyeongjun; Park, Jihwan; Kim, Suhwan

被引用次数: 14

2019

A 2-MHz BW 82-dB DR Continuous-Time Delta-Sigma Modulator With a Capacitor-Based Voltage DAC for ELD Compensation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Susie; Na, Seung-In; Yang, Youngtae; Kim, Suhwan

被引用次数: 5

2018

An Uncooled Microbolometer Infrared Imager With a Shutter-Based Successive-Approximation Calibration Loop

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Park, Yujin; Yun, Junghee; Park, Dongchul; Kim, Sangwoo; Kim, Suhwan

被引用次数: 6

2018

A Stereo Audio Delta-Sigma DAC with 40-kHz Bandwidth and 103-dB SNR

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Yang, Han; Cho, Jun Soo; Park, Yujin; Kim, Hyunjong; Kim, Suhwan

被引用次数: 0

2018

MATLAB/Simulink Pulse-echo Ultrasound System Simulator with Electrical Impedance Matching

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Shin, Sangmin; Kim, Taehoon; Kim, Suhwan

被引用次数: 1

2018

A 1.2-V 108.9-dB A-Weighted DR 101.4-dB SNDR Audio $\Sigma\Delta$ ADC Using a Multi-Rate Noise-Shaping Quantizer

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Cho, Jun Soo; Rhee, Cyuyeol; Kim, Susie; Yang, Youngtae; Jun, Jaehoon; Kim, Suhwan; Lee, Hyunjoong

被引用次数: 6

2018

A 4266 Mb/s/pin LPDDR4 Interface With An Asynchronous Feedback CTLE and An Adaptive 3-Step Eye Detection Algorithm for Memory Controller

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Mino; Chae, Joo-Hyung; Choi, Sungphil; Hong, Gi-Moon; Ko, Hyeongjun; Jeong, Deog-Kyoon; Kim, Suhwan

被引用次数: 6

2018

A 2.1-Gb/s 12-Channel Transmitter With Phase Emphasis Embedded Serializer for 55-in UHD Intra-Panel Interface

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Jihwan; Chae, Joo-Hyung; Jeong, Yong-Un; Lee, Jae-Whan; Kim, Suhwan

被引用次数: 10

2018

A 3.2 Gb/s 16-Channel Transmitter for Intra-Panel Interfaces, With Independently Controllable Output Swing, Common-mode Voltage, and Equalization

IEEE ACCESS

Chae, Joo-Hyung; Kim, Mino; Hong, Gi-Moon; Park, Jihwan; Kim, Suhwan

被引用次数: 2

2018

0.11-2.5 GHz All-digital DLL for Mobile Memory Interface with Phase Sampling Window Adaptation to Reduce Jitter Accumulation

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Chae, Joo-Hyung; Kim, Mino; Hong, Gi-Moon; Park, Jihwan; Ko, Hyeongjun; Shin, Woo-Yeol; Chi, Hankyu; Jeong, Deog-Kyoon; Kim, Suhwan

被引用次数: 6

2017

A 4x32-Channel Neural Recording System for Deep Brain Stimulation Systems

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Susie; Na, Seung-In; Yang, Youngtae; Kim, Hyunjong; Kim, Taehoon; Cho, Jun Soo; Kim, Jinhyung; Chang, Jin Woo; Kim, Suhwan

被引用次数: 6

2017

A Sub-1.0-V On-Chip CMOS Thermometer With a Folded Temperature Sensor for Low-Power Mobile DRAM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Hyunjoong; Shim, Daeyong; Rhee, Cyuyeol; Kim, Mino; Kim, Suhwan

被引用次数: 4

2016

Column readout circuit with dual integration CDS for infrared imagers

IEICE ELECTRONICS EXPRESS

Park, Yujin; Yang, Han; Ahn, Jaehyun; Kim, Suhwan

被引用次数: 1

2016

Estimating Non-Ideal Effects within a Top-Down Methodology for the Design of Continuous-Time Delta-Sigma Modulators

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Na, Seung-in; Kim, Susie; Yang, Youngtae; Kim, Suhwan

被引用次数: 2

2016

Design of a Two-Stage Driver for LED MR16 Retrofit Lamps Compatible with Electronic Transformers

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Yim, Sungwon; Lee, Hyongmin; Lee, Bongjin; Kang, Kyucheol; Kim, Suhwan

被引用次数: 1

2016

Differentiating ASK Demodulator for Contactless Smart Cards Supporting VHBR

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Hyongmin; Kim, Jisung; Ha, Dongwoo; Kim, Taehoon; Kim, Suhwan

被引用次数: 29

2015

1.2 V 10-bit 75 MS/s Pipelined ADC With Phase-Dependent Gain-Transition CDS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Woo, Jong-Kwan; Lee, Hyunjoong; Kim, Hwi-Cheol; Jeong, Deog-Kyoon; Kim, Suhwan

被引用次数: 6

2014

A PVT-compensated 2.2 to 3.0 GHz Digitally Controlled Oscillator for All-Digital PLL

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kavala, Anil; Bae, Woorham; Kim, Sungwoo; Hong, Gi-Moon; Chi, Hankyu; Kim, Suhwan; Jeong, Deog-Kyoon

被引用次数: 8

2014

A comparator-based cyclic analog-to-digital converter with multi-level input tracking boosted preset voltage

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Woo, Jong-Kwan; Kim, Taehoon; Kim, Suhwan

被引用次数: 1

2014

A Process-Variation-Tolerant On-Chip CMOS Thermometer for Auto Temperature Compensated Self-Refresh of Low-Power Mobile DRAM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shim, Daeyong; Jeong, Hyunsik; Lee, Hyunjoong; Rhee, Cyuyeol; Jeong, Deog-Kyoon; Kim, Suhwan

被引用次数: 31

2013

4-Slot, 8-Drop Impedance-Matched Bidirectional Multidrop DQ Bus With a 4.8-Gb/s Memory Controller Transceiver

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Shin, Woo-Yeol; Hong, Gi-Moon; Lee, Hyongmin; Han, Jae-Duk; Park, Kyu-Sang; Lim, Dong-Hyuk; Kim, Sunkwon; Shim, Daeyong; Chun, Jung-Hoon; Jeong, Deog-Kyoon; Kim, Suhwan

被引用次数: 8

2013

A highly selective chemical sensor array based on nanowire/nanostructure for gas identification

SENSORS AND ACTUATORS B-CHEMICAL

Hu, Yushi; Lee, Hyunjoong; Kim, Suhwan; Yun, Minhee

被引用次数: 30

2013

A 10-Mbps 0.8-pJ/bit Referenceless Clock and Data Recovery Circuit for Optically Controlled Neural Interface System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Sunkwon; Woo, Jong-Kwan; Shin, Woo-Yeol; Hong, Gi-Moon; Lee, Hyongmin; Lee, Hyunjoong; Kim, Suhwan

被引用次数: 4

2013

Bitline Techniques With Dual Dynamic Nodes for Low-Power Register Files

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Singh, Rahul; Hong, Gi-Moon; Kim, Suhwan

被引用次数: 8

2013

Detection and Identification of Breast Cancer Volatile Organic Compounds Biomarkers Using Highly-Sensitive Single Nanowire Array on a Chip

JOURNAL OF BIOMEDICAL NANOTECHNOLOGY

Xu, Yiwen; Lee, Hyunjoong; Hu, Yushi; Huang, Jiyong; Kim, Suhwan; Yun, Minhee

被引用次数: 50

2013

Integrating Metal-Oxide-Decorated CNT Networks with a CMOS Readout in a Gas Sensor
SENSORS

Lee, Hyunjoong; Lee, Sanghoon; Kim, Dai-Hong; Perello, David; Park, Young June; Hong, Seong-Hyeon; Yun, Minhee; Kim, Suhwan

被引用次数: 15

2012

Static-switching pulse domino: A switching-aware design technique for wide fan-in dynamic multiplexers

INTEGRATION-THE VLSI JOURNAL

Singh, Rahul; Hong, Gi-Moon; Kim, Mino; Park, Jihwan; Shin, Woo-Yeol; Kim, Suhwan

被引用次数: 5

2012

Power-Gating Noise Minimization by Three-Step Wake-Up Partitioning

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Singh, Rahul; Woo, Jong-Kwan; Lee, Hyunjoong; Kim, So Young; Kim, Suhwan

被引用次数: 9

2012

A 1.0-4.0-Gb/s All-Digital CDR With 1.0-ps Period Resolution DCO and Adaptive Proportional Gain Control

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, Heesoo; Kim, Deok-Soo; Oh, Do-Hwan; Kim, Suhwan; Jeong, Deog-Kyoon

被引用次数: 68

2011

A Delta-Sigma Interface Circuit for Capacitive Sensors With an Automatically Calibrated Zero Point

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Shin, Dong-Yong; Lee, Hyunjoong; Kim, Suhwan

被引用次数: 81

2011

A Low-Cost and Low-Power Time-to-Digital Converter Using Triple-Slope Time Stretching

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Manho; Lee, Hyunjoong; Woo, Jong-Kwan; Xing, Nan; Kim, Min-Oh; Kim, Suhwan

被引用次数: 21

2011

Optical Failure Analysis Technique in Deep Submicron CMOS Integrated Circuits

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Sunkwon; Lee, Hyongmin; Lee, Hyunjoong; Woo, Jong-Kwan; Cheon, Junho; Kim, Hwan-Yong; Park, Young June; Kim, Suhwan

被引用次数: 0

2011

Improving the Accuracy of Capacitance-to-Frequency Converter by Accumulating Residual Charges

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Shin, Dong-Yong; Lee, Hyunjoong; Kim, Suhwan

被引用次数: 15

2011

A 0.3-1.4 GHz All-Digital Fractional-N PLL With Adaptive Loop Gain Controller

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Deok-Soo; Song, Heesoo; Kim, Taeho; Kim, Suhwan; Jeong, Deog-Kyoon

被引用次数: 48

2010

A 14.6 ps Resolution, 50 ns Input-Range Cyclic Time-to-Digital Converter Using Fractional Difference Conversion Method

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xing, Nan; Woo, Jong-Kwan; Shin, Woo-Yeol; Lee, Hyunjoong; Kim, Suhwan

被引用次数: 27

2010

A Fast-Acquisition PLL using Split Half-Duty Sampled Feedforward Loop Filter

IEEE TRANSACTIONS ON CONSUMER ELECTRONICS

Shin, Woo-Yeol; Kim, Manho; Hong, Gi-Moon; Kim, Suhwan

被引用次数: 5

2010

Biosensor system-on-a-chip including CMOS-based signal processing circuits and 64 carbon nanotube-based sensors for the detection of a neurotransmitter

LAB ON A CHIP

Lee, Byung Yang; Seo, Sung Min; Lee, Dong Joon; Lee, Minbaek; Lee, Joohyung; Cheon, Jun-Ho; Cho, Eunju; Lee, Hyunjoong; Chung, In-Young; Park, Young June; Kim, Suhwan; Hong, Seunghun

被引用次数: 37

2010

Charge Amplifier With an Enhanced Frequency Response for SPM-Based Data Storage

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Hyunjoong; Woo, Jong-Kwan; Nam, Hyo-Jin; Jin, Won-Hyeog; Jeong, Moongi; Kim, Young-Sik; Shin, Jin-Koog; Kim, Suhwan

被引用次数: 5

2010

A High-Speed Range-Matching TCAM for Storage-Efficient Packet Classification

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Young-Deok; Ahn, Hyun-Seok; Kim, Suhwan; Jeong, Deog-Kyoon

被引用次数: 26

2009

A Low-Power Register File with Dual-Vt Dynamic Bit-Lines driven by CMOS Bootstrapped Circuit

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Hyoung-Wook; Lee, Hyunjoong; Woo, Jong-Kwan; Shin, Woo-Yeol; Kim, Suhwan

被引用次数: 0

2009

A Reduced-Swing Voltage-Mode Driver for Low-Power Multi-Gb/s Transmitters

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Song, Heesoo; Kim, Suhwan; Jeong, Deog-Kyoon

被引用次数: 11

2009

High-Speed 10-bit LCD Column Driver with a Split DAC and a Class-AB Output Buffer

IEEE TRANSACTIONS ON CONSUMER ELECTRONICS	
Woo, Jong-Kwan; Shin, Dong-Yong; Jeong, Deog-Kyoon; Kim, Suhwan	
被引用次数: 24	2009
Reducing ground-bounce noise and stabilizing the data-retention voltage of power-gating structures	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Kim, Suhwan; Choi, Chang Jun; Jeong, Deog-Kyoon; Kosonocky, Stephen V.; Park, Sung Bae	
被引用次数: 29	2008
A multi-mode power Gating structure for low-voltage deep-submicron CMOS ICs	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Kim, Suhwan; Kosonocky, Stephen V.; Knebel, Daniel R.; Stawiasz, Kevin; Papaefthymiou, Marios C.	
被引用次数: 53	2007
Charge-recovery computing on silicon	
IEEE TRANSACTIONS ON COMPUTERS	
Kim, S; Ziesler, CH; Papaefthymiou, MC	
被引用次数: 54	2005
A true single-phase energy-recovery multiplier	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Kim, S; Ziesler, CH; Papaefthymiou, MC	
被引用次数: 38	2003
Low-power circuits and technology for wireless digital systems	
IBM JOURNAL OF RESEARCH AND DEVELOPMENT	
Kosonocky, SV; Bhavnagarwala, AJ; Chin, K; Gristede, GD; Haen, AM; Hwang, W; Ketchen, MB; Kim, S; Knebel, DR; Warren, KW; Zyuban, V	
被引用次数: 15	2003
True single-phase adiabatic circuitry	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Kim, S; Papaefthymiou, MC	
被引用次数: 74	2001

学术合作

